

ACTIVE POWER CONDITIONERS FOR HARMONICS

Read on to know about a power conditioner that can sense power load requirements and handle fluctuations every 200 microseconds



Inside view of iCon 210A

■ JALAJA RAMANUNNI

Business organisations are constantly looking for efficient energy solutions, especially with the government regulations getting stricter, and rising costs involved with power usage.

P2 Power Solutions, a Noida-based company has developed iCon—an active power conditioner—which it claims will help in power factor (PF) correction, harmonics filtering, load

balancing, neutral protection and bring a facility's energy consumption down by at least 50 per cent.

Problems faced in normal scenario

iCon takes care of power quality issues like poor PF, harmonics in the supply system, unbalanced loads and over-current in the neutrals. "The normal electricity supply we get has a frequency of 50 hertz. However, due to

some heavy loads and equipment like motors, drives, UPS, lighting loads and other non-linear loads, waveforms of other frequencies which are odd multiples of 50, e.g., 150, 250, 350 or 550 hertz, are introduced in the supply system. As a result, harmonics currents cause over-heating of the cables and transformers, random tripping of circuit breakers, or even damage to other machines connected to the system. Traditionally used capacitor-banks for PF improvement further intensify the ill-effects of the harmonic currents. This causes the maintenance costs, energy bills and other overhead costs to substantially rise up," illustrates Rajeev Pandey, head, product development.

Plugging a power conditioner can help to produce a clean flow of regulated current to the equipment. Pandey feels that iCon is different from other power conditioners. "What sets iCon apart from other power conditioners is its ability to sense the load requirement every 200 microseconds, and hence, manage severe fluctuations in the power distributed to equipment. Meanwhile, factor correction system (RTFPC) takes a response time of three to five milliseconds and capacitor-banks controlled by relays take up to three seconds to sense the load and then do the necessary correction. Therefore these traditional solutions are comparatively much slower in managing the power fluctuations. Further, the ability of iCon to handle the Holy Grails of power related issues—low PF, harmonics, unbalanced